

Module 3 Homework

 Published

Java Class and Objects Homework

Total Points: 100

Objective: The objective of this homework assignment is to assess your understanding of Java classes and objects. You will create two Java classes and utilize objects to demonstrate your knowledge.

Class 1: Student

Create a Java class named "Student" with the following attributes and methods:

1. Attributes:

- `name` (String): the name of the student.
- `rollNumber` (int): a unique identifier for the student.
- `marks` (an array of integers): representing the marks of the student in various subjects. Assume there are five subjects.

2. Methods:

- A constructor to initialize `name`, `rollNumber`, and `marks`.
- `getTotalMarks()`: a method to calculate and return the total marks of the student.
- `getAverageMarks()`: a method to calculate and return the average marks of the student.
- `getGrade()`: a method to return the grade based on the average marks. Use the following grading system:
 - A: 90-100
 - B: 80-89
 - C: 70-79
 - D: 60-69
 - F: Below 60

Class 2: School

Create another Java class named "School" with the following attributes and methods:

1. Attributes:

- `students` (an array of Student objects): representing a list of students in the school.

2. Methods:

- A constructor to initialize the `students` array.
- `getStudentWithHighestAverage()`: a method to find and return the student with the highest average marks in the school.
- `getStudentWithLowestAverage()`: a method to find and return the student with the lowest average marks in the school.

Grading Rubric:

• Class 1: Student (40 points)

- Attribute declaration and initialization: 5 points
- Constructor implementation: 10 points
- `getTotalMarks()` method: 5 points
- `getAverageMarks()` method: 10 points
- `getGrade()` method: 10 points

• Class 2: School (40 points)

- Attribute declaration and initialization: 5 points
- Constructor implementation: 10 points
- `getStudentWithHighestAverage()` method: 15 points
- `getStudentWithLowestAverage()` method: 10 points

• Documentation and Comments (10 points)

- Clear and concise comments: 5 points
- Properly formatted code: 5 points

• Bonus (10 points)

- Implement additional functionality, such as sorting the students by average marks, and demonstrate its usage in the main program.

- **Total Points (100 points)**

Note:

- You should create a main program to demonstrate the functionality of the Student and School classes.
- Properly document your code with comments and provide sample output from your program.
- Ensure your code is well-organized, follows best practices, and is free of syntax errors.

Points 100

Submitting a file upload

Due	For	Available from	Until
Oct 4	Everyone	-	Oct 22 at 11:59pm